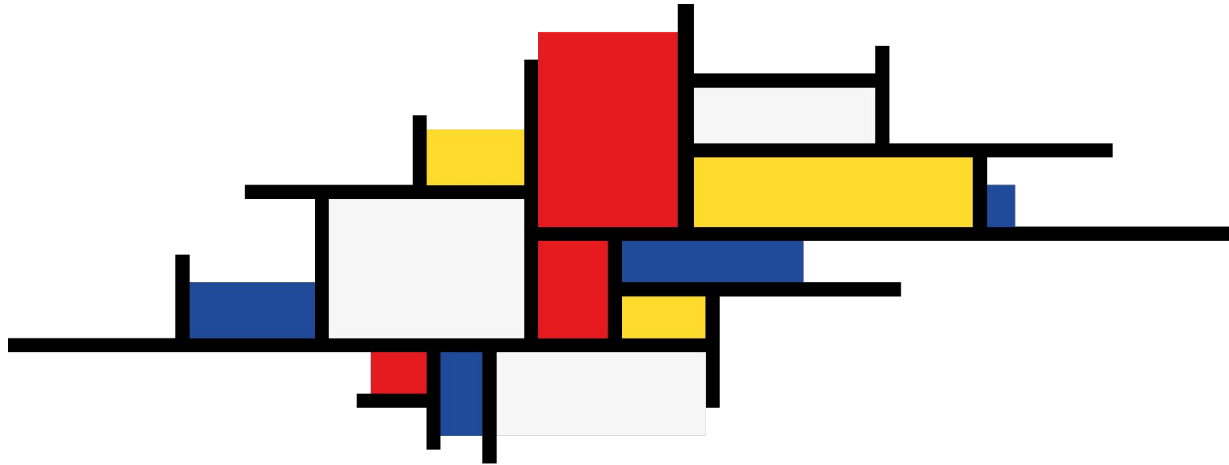
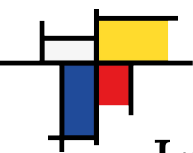


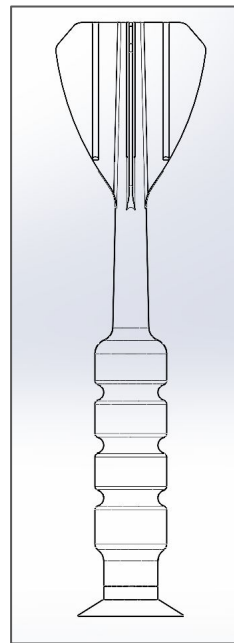
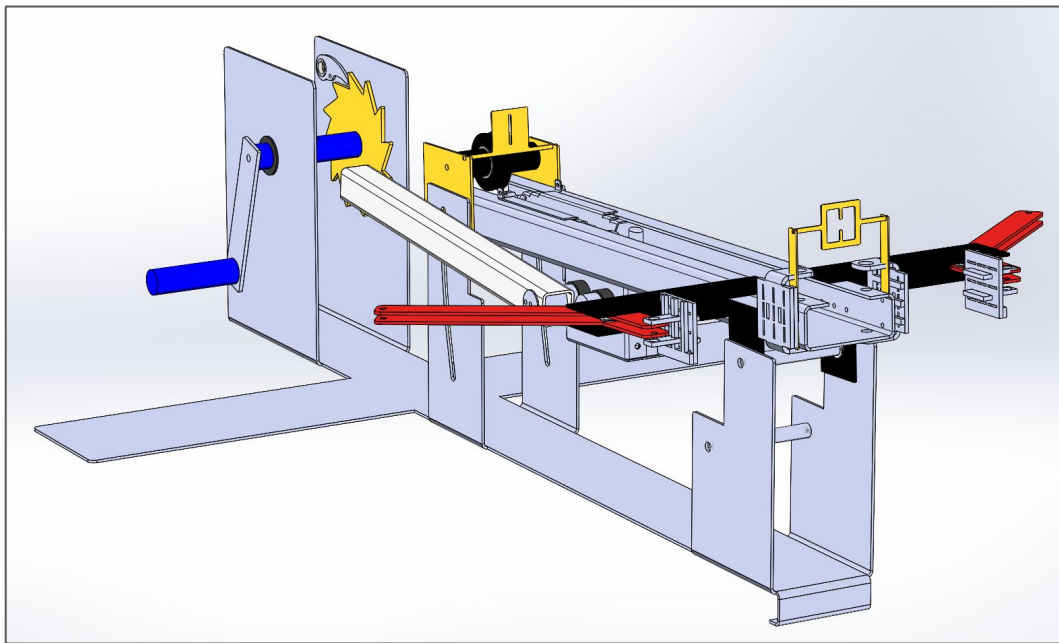
The Twang.

The White Lightning





Introduction



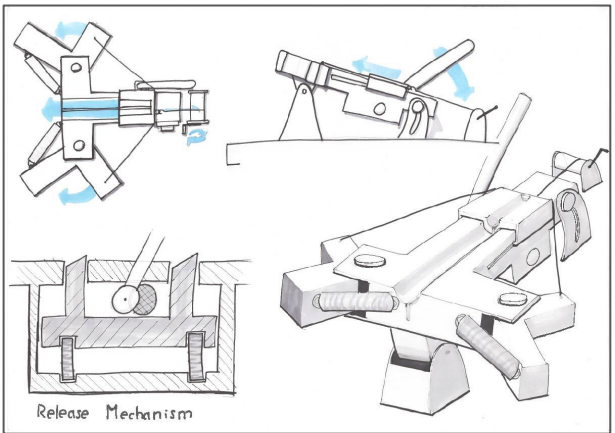


Features, Process, Underpinning

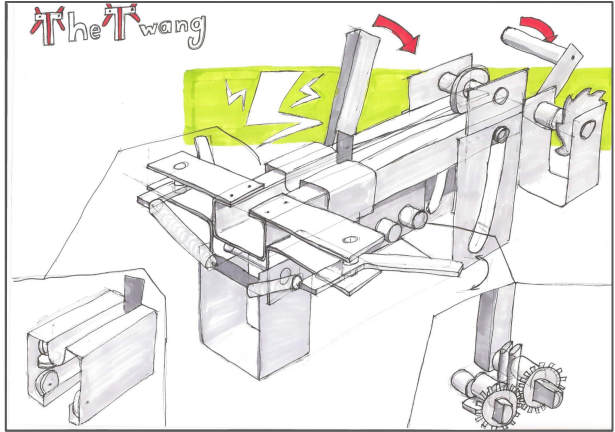
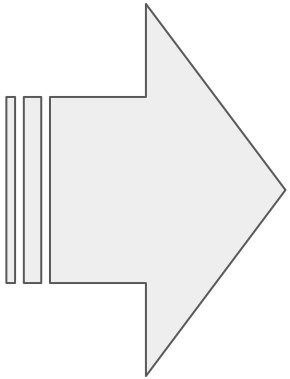
- ❖ Where did we start?
 - First concept sketch
 - Final concept sketch
- ❖ Where did we go?
 - CAD model
- ❖ Design justification



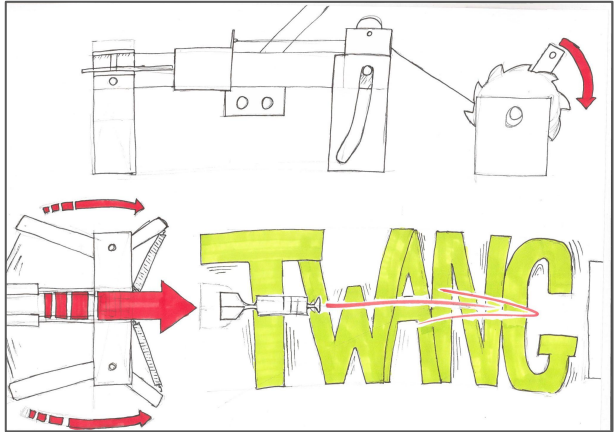
Concept Sketches



Initial concept drawing



Final concept drawing

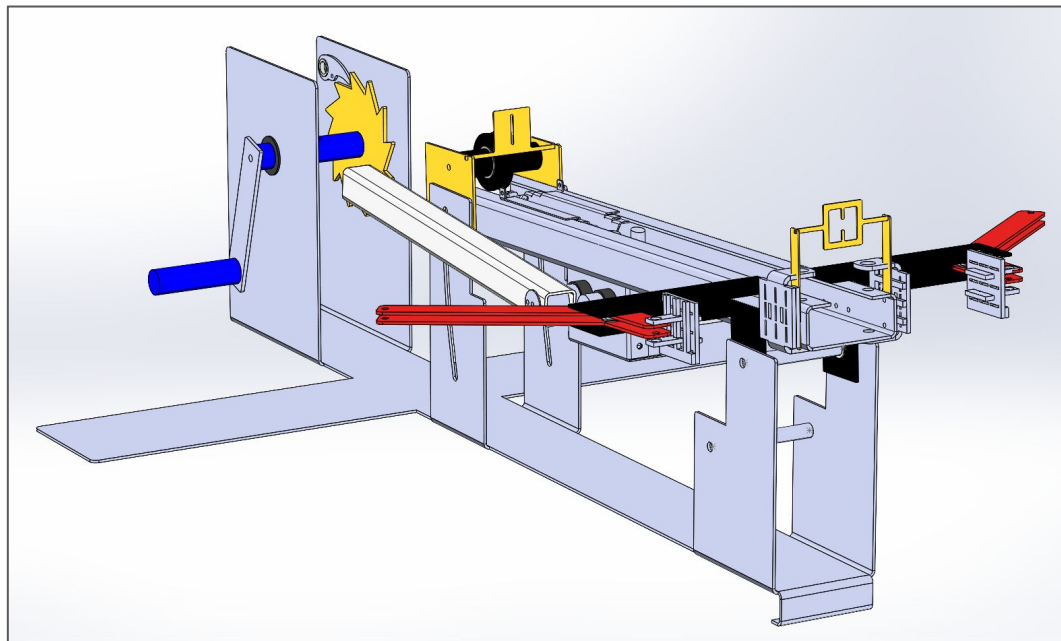




Assemblies

Total Assembly

- ❖ Baseplate
- ❖ Winch
- ❖ Frame
 - Cart
 - Spring Modules
 - T-Shape
 - Visors
 - Pulley
 - Release Mechanism
 - Casing
 - Pinplate
 - Acylindrical Bars





Assemblies

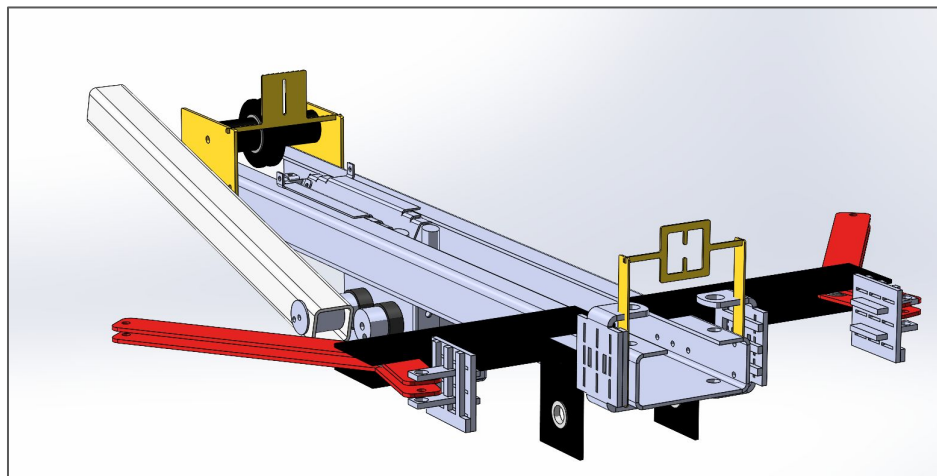
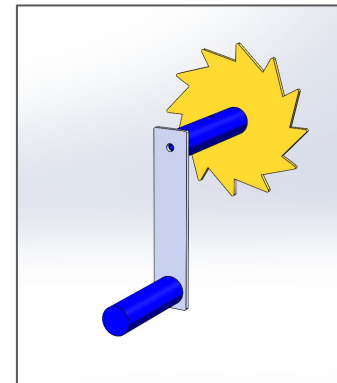
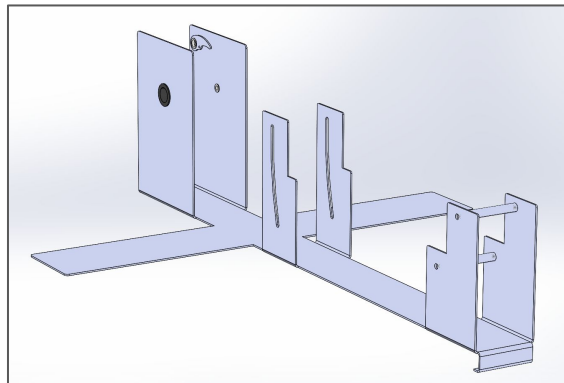
Total Assembly

❖ **Baseplate**

❖ **Winch**

❖ **Frame**

- Cart
- Spring Modules
- T-Shape
 - Visors
 - Pulley
 - Release Mechanism
 - Casing
 - Pinplate
 - Acylindrical Bars

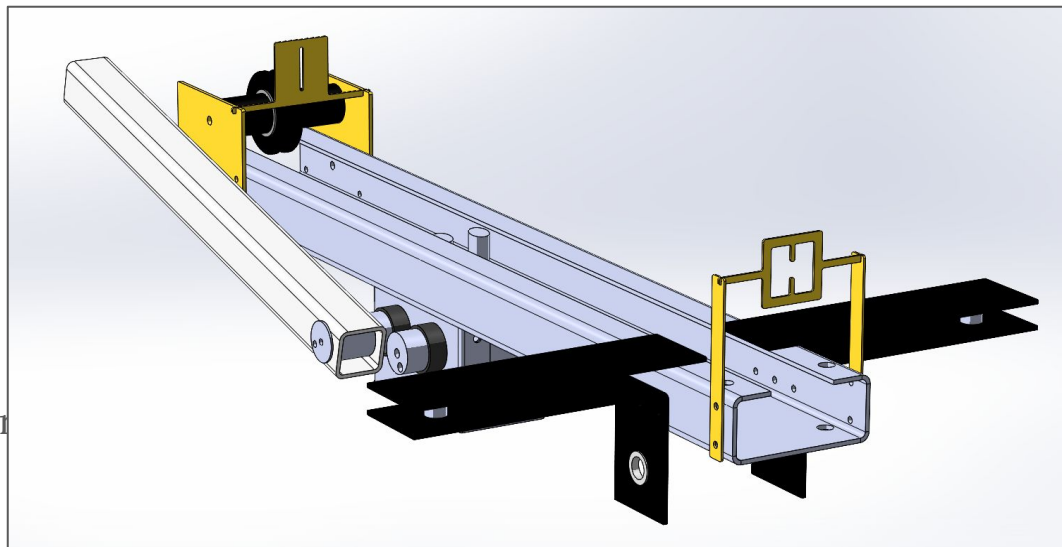
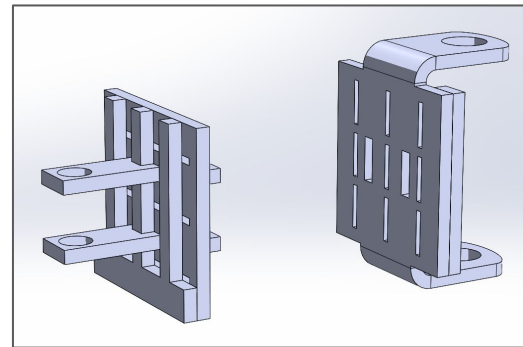
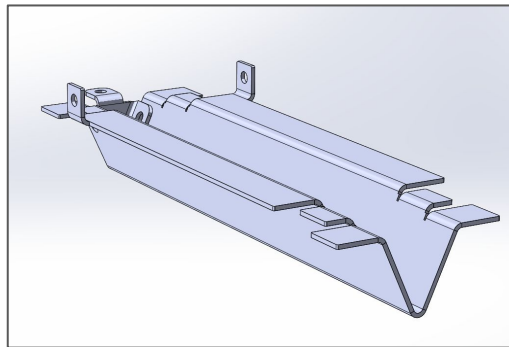




Assemblies

Total Assembly

- ❖ Baseplate
- ❖ Winch
- ❖ Frame
 - **Cart**
 - **Spring Modules**
 - **T-Shape**
 - Visors
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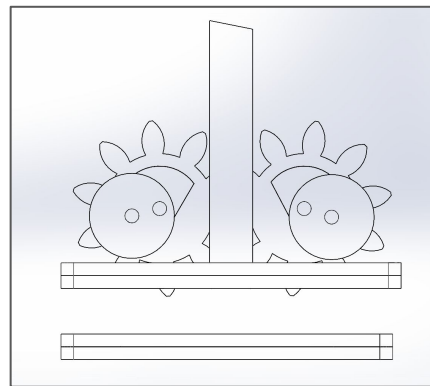
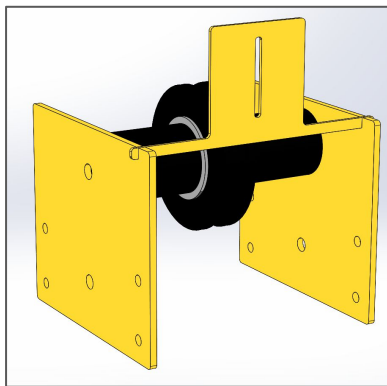
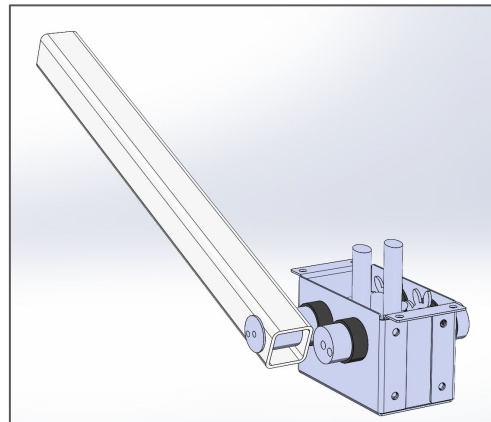
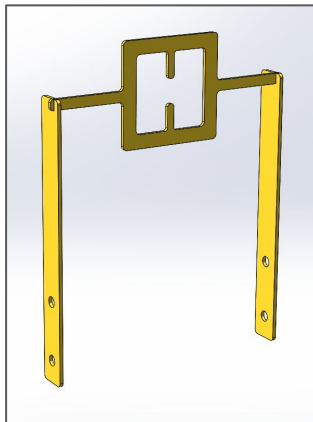




Assemblies

Total Assembly

- ❖ Baseplate
- ❖ Winch
- ❖ Frame
 - Cart
 - Spring Modules
 - T-Shape
 - **Visor**
 - **Pulley**
 - **Release Mechanism**
 - Casing
 - Pinplate
 - Acylindrical Bars

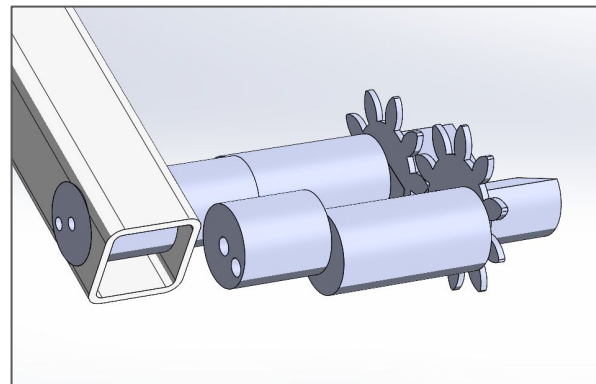
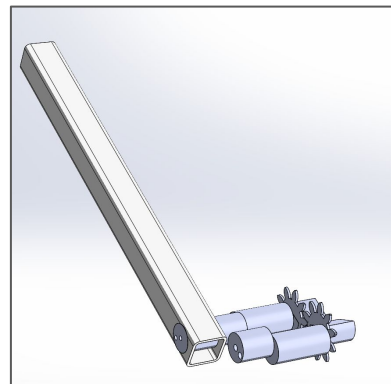
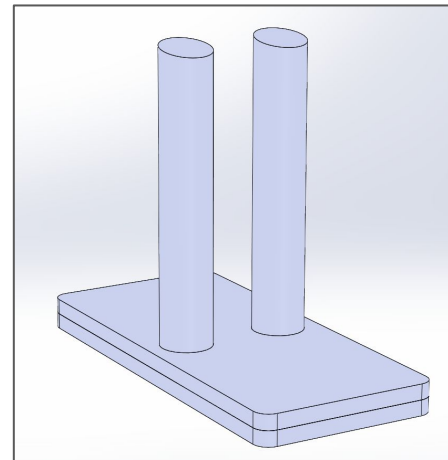
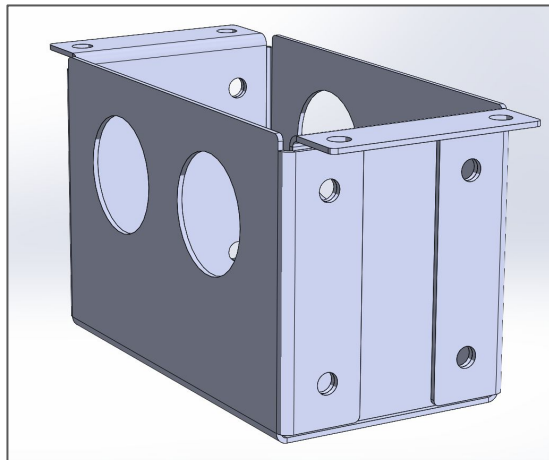




Assemblies

Total Assembly

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 - **Casing**
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 - **Acyindrical Bars**





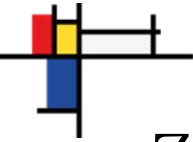
Working, demonstration, tests

- ❖ How does the device work?
- ❖ Close look at each part
 - Style
- ❖ Testing the device + Videos
 - Dart matches



How does the device work?

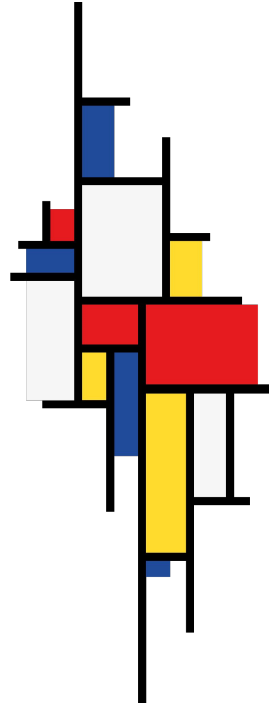
- ❖ Fastening
- ❖ Aiming
- ❖ Charging
- ❖ Releasing



Zooming in

- ❖ How did each part actually turn out?
 - Did any flaws negatively impact the device?
 - Style
 - Some amateur painting → cool effect

Videos





Testing

❖ Hangar

- Only a few test shots
 - Spring modules
 - Fewer, but stronger springs
 - Hook
 - Joinings broke → Reinforced the small T and cart
 - Destroying itself
- Results
 - Requirements met, limited testing
 - Only for 4 meters
 - Carriable → not on a bike (don't try it)

❖ Dart matches

- What went wrong?!
 - Too little testing
 - Deformed materials (FMEA), foam padding



Failings, evaluation, improvements

Small improvements



Testing

Testing the products functionality

Discovering failings

Discovering the things that fail the requirements.

Evaluation

Evaluate prototype, based on the testing.

Improvements

Improvements that still need to be made.



Testing results

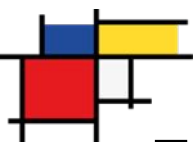
Tested aspect	Results(avg.)
Time putting dartshooter on the table.	0:19
Time shooting 3 darts	1:35
Shot accuracy(darts hit and stick)	
4m	2
6m	0
8m	0
Is the dartshooter carryable by one person?	Yes
Is the dartshooter operable by one person.	Yes
Answer on question: Is it hard to pull back the cart?	No



Discovering failings

When the prototype was tested problems arose:

- ❖ Darts could not reach the board from 8 meters.
- ❖ Cart went over the Small T top.
- ❖ Cart was destroying itself.
- ❖ Lever of the release mechanism was hitting the table.
- ❖ Release pins did not release the cart
- ❖ Baseplate-spacers, were too long.
- ❖ Baseplate was bending, when the cart was pulled back.
- ❖ Rivets between the big and small T broke.
- ❖ Lever of the release mechanism was falling down, so it was very hard to operate the dart shooter on your own.
- ❖ Small T top could not handle the force of the cart.



Evaluation

Tested aspect	Results(avg.)	Requirement
Time putting dartshooter on the table.	0:19	Met
Time shooting 3 darts	1:35	Met
Shot accuracy(darts hit and stick)		Not met
4m	2	
6m	0	
8m	0	
Is the dartshooter carryable by one person?	Yes	Met
Is the dartshooter operable by one person.	Yes	Met
Answer on question: Is it hard to pull back the cart?	No	Met



Improvements

Requirements not met:

- ❖ *“The dart shooter shall shoot the darts on the blazon 4-8 meters away.”*
- ❖ *“The dart shooter shall be operable by one person.”*

Other:

- ❖ Cart was slowly destroying itself.
- ❖ Safer, a lot of moving parts were not shielded.



Final remarks?

